

RegisterMachine N, D, and L Experiment Tables

All accuracies are percentages. Trace-step accuracy and colon rate are omitted.

Sweep	Controlled variable	Fixed configuration
N	Training examples: 25k, 50k, 100k	D=16, L=2, seed=2001, 8,000 updates
D	Execution depth: 4, 8, 12, 16	N=100k, L=2, seed=2001, 8,000 updates
L	Transformer blocks: 2, 4, 6	N=100k, D=16, seed=2001, 8,000 updates

Data provenance: N and D=4 values are from the uploaded CSV runs; D=8/12/16 and L=2/4/6 use the latest complete runs shared in the terminal logs.

N sweep - training-set size

Fixed: RegisterMachine depth D=16, Transformer depth L=2, seed=2001, 8,000 optimizer updates.

rho	25k		50k		100k	
	Answer	Exact trace	Answer	Exact trace	Answer	Exact trace
0.0	6.25	0.00	5.90	0.00	6.65	0.00
0.1	6.65	0.20	8.60	0.40	8.80	0.50
0.2	29.50	20.05	33.40	25.05	30.85	21.30
0.3	65.50	59.95	58.60	52.40	68.90	63.90
0.4	77.35	73.10	87.40	85.75	88.85	86.90
0.5	88.85	86.85	92.25	91.25	94.25	93.40
0.6	91.20	89.85	96.55	96.05	97.95	97.55
0.7	97.45	97.05	97.50	97.25	97.70	97.40

D sweep - execution depth

Fixed: N=100,000, Transformer depth L=2, seed=2001, 8,000 optimizer updates.

rho	4		8		12		16	
	Answer	Exact trace	Answer	Exact trace	Answer	Exact trace	Answer	Exact trace
0.0	72.25	0.00	6.65	0.00	5.75	0.00	6.45	0.00
0.1	71.60	7.40	22.75	9.35	10.05	1.15	7.65	0.30
0.2	60.80	39.85	67.25	57.70	38.45	28.90	28.70	18.45
0.3	81.95	75.30	87.60	84.60	77.90	75.25	66.35	61.60
0.4	95.20	94.15	93.30	92.05	91.90	91.00	89.20	87.75
0.5	97.45	97.10	97.75	97.35	95.65	94.80	96.05	95.40
0.6	98.90	98.75	99.20	98.95	97.45	97.20	97.60	97.30
0.7	98.25	98.10	98.95	98.55	99.45	99.40	98.75	98.50

L sweep - model depth / capacity

Fixed: N=100,000, RegisterMachine depth D=16, seed=2001, 8,000 optimizer updates.

rho	2		4		6	
	Answer	Exact trace	Answer	Exact trace	Answer	Exact trace
0.0	6.55	0.00	6.65	0.00	6.65	0.00
0.1	8.40	0.40	43.95	39.50	39.95	35.05
0.2	29.10	19.00	89.50	88.45	87.20	86.15
0.3	72.60	69.00	78.85	76.65	93.55	93.10
0.4	89.35	87.50	92.45	91.40	98.65	98.50
0.5	94.25	93.50	98.80	98.70	99.10	99.00
0.6	97.80	97.25	99.10	99.10	98.35	98.20
0.7	98.35	98.30	98.75	98.55	99.75	99.75